

Data Management is People Management:

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Abstract

Archaeological projects establish organizational structures to coordinate labour and information systems to pool the data produced from that labour. These data streams can then be applied toward the production and publication of analytical findings. However, these information commons that are generated through collective efforts are not egalitarian; project communities establish practical expectations for how people in different positions should engage with archaeological information based on their roles. Contributing to and extracting from the commons is therefore scaffolded by a diverse yet converging series of commitments to a collective enterprise, which are instilled through participation in a community of practice.

Through an abductive qualitative data analysis based on recorded observations, interviews, and documents collected from two cases, this paper highlights how digital systems designed to direct the flow of information do so via the coordination of labour and the strategic arrangement of human and object agency. Specifically, it highlights how the information systems that archaeologists rely on to generate records, internal reports, published papers and integrated datasets reify, reinforce, and sometimes challenge entrenched power structures and divisions of labour. By characterizing the documentary media that archaeologists use to capture, organize and share knowledge as tools that facilitate controlled communication and which govern how people may contribute to and access collectively-maintained knowledge, this paper re-casts the formation of data commons as a social rather than a technical enterprise.

Note

This is a pre-review preprint of a paper presented at the 2nd Workshop in the [Exploring the Layers of Digital Archaeological Practice Series](#), whose theme for 2025 was [Data and Technology Politics in Archaeology](#). The workshop was hosted at the Norwegian Institute at Athens from December 3-4, 2025. This content is also available at zackbatist.info/data-tech-poli-arch for a better reading experience.

Introduction

- [1] Archaeological projects are inherently collaborative, in that they bring teams of people with a variety of unique outlooks and experiences to examine material assemblages of common interest. Archaeologists apply a multitude of tools and methods, in sequence and in tandem and across different kinds of work settings, to produce rich and heterogeneous data about their engagements with the archaeological record. They adhere to professional norms and expectations to organize the products of their collective labour, and rely on a combination of digital and analog devices and protocols to pool the outcomes of their respective tasks.

- [2] Advanced computational systems, especially those that integrate data from across the span of work occurring within archaeological projects, have helped overcome technical barriers obstructing the free flow of information. These systems have undoubtedly unlocked enhanced analytical protocols that support robust understanding of the past. However, data integration is much more than a technical process; it also involves considerable collaborative labour to facilitate commensurability of research outcomes and to ensure that all parties effectively contribute to common goals.
- [3] In this paper I make clear associations between project and data management, and how they establish means of maintaining controlled information resources. It therefore draws attention to the series of collaborative commitments and broader social contexts in which archaeologists embed their work. Specifically, the paper examines how locally-circumscribed practices are re-framed as generic processes as part of attempts to streamline the assembly and dissemination of data, and how this permeates across project systems.

Background

- [4] Archaeological data are records that encode objects' characteristics, properties or features in ways that render meaningful aspects communicable among actors who share common points of reference.¹ In effect, data identify, record and format information about objects or phenomena and their relevant properties, which enables knowledge about them to circulate and form the basis for extended avenues of knowledge production. It is therefore helpful to frame data sharing as an act of discursive communication, whereby data are documentary records that enable research to be extended across time and place, and to be carried out in a collaborative manner. The media upon which data are inscribed (e.g. databases, registries, and online catalogues) enable direct experiences with objects and phenomena of interest to be shared and acted upon in alternative research contexts. This view also encourages consideration of data sharing within the internal purview of independent projects. Overall, thinking of data sharing as acts of communication between actors with overlapping perspectives and expectations draws out a greater appreciation for the ecologies of knowledge involved in the creation and curation of archaeological data, and of tensions between the promise of aggregated open data and the lived realities of professional practice (Batist 2023).
- [5] The past decade has produced a flood of research that critically examines archaeological information systems along those lines. Although there is a long history of reflexive research on the social and collaborative nature of archaeological data, and there are too many scattered instances of such work being applied to examine specific areas of archaeological practice, a few prior works stand out as particularly relevant in the context of the present study. Hacıgüzeller, Taylor, and Perry (2021) looked at the dominance of structured data and highlighted certain myths regarding what structured data is generally thought to afford. This complements Huggett (2022) who articulated some means through our imaginaries of data as concrete and objective representations are manifested. These perspectives align with a theoretical framework offered by (Dallas 2015, 2016), who sought to re-frame all epistemic tasks as series of pragmatic, dialectic exchanges, whereby work is informed by internalized anticipation of future extensions of the task's outcomes and of reconciliation of the prior decisions that shaped circumstances under which the task is being undertaken.
- [6] Others have drawn critical attention to mechanisms through which projects and institutions seek to control these epistemic acts that occur in archaeological data collection and management. Eddisford and Morgan (2019) draw from Star (1993) who examined the means through which recording sheets are applied to control the means through which data are collected in field-based research. This relates to research by Yarrow (2008),

¹These may include both human and non-human actors.

Mickel (2021) and Thorpe (2012) who identified paternalistic attitudes concerning documentation practices in pedagogical, neocolonial and commercial contexts, respectively. Batist et al. (2021) extended this line of inquiry by noting how the use of digital systems are often used in a way that reduces crafty and creative applications; as such, the tools impose order on research practice, leading the archaeologists to manage their projects and educate their students in ways that fulfill the tools' needs.

Methods and Data

- [7] The findings presented here derive from the author's dissertation (Batist 2023), which produced three other papers utilizing overlapping source material (Batist 2024, 2025b, 2025a). Refer to the methods and data presented in those works for further methodological details.

Cases

- [8] This paper articulates some systemic relations that scaffold archaeologists' actions and attitudes at two independent archaeological projects, which serve as cases. The cases are not the subjects of inquiry, and instead represent discrete instances that share common reference to the overall research themes (Stake 2006). So while the cases are certainly not representative of the whole archaeological discipline, they do enable me to reveal some underappreciated systemic concerns that underlie data management. As such, the findings are informed by the behaviours and attitudes expressed to me by those who agreed to participate, and by my own standpoint as a scholar of the culture and practice of science and of the media and infrastructures that support data sharing, integration and reuse.
- [9] Both cases were research projects located in southern Europe and directed by professors affiliated with North American universities. I contributed to Case A for several years as a database manager, which provided me a privileged outlook on the systems set up to integrate and draw value from data. My involvement as a participant observer occurred over three years, from 2017 to 2019, and involved visiting the project during its summer field seasons to observe archaeological practices and to interview selected participants. I also held interviews throughout the "off season". My continual and longer-term participation at Case A afforded greater understanding of the intricate social relations that developed over time, and enabled me to examine how certain methods and attitudes evolved over successive field seasons. This helped me account for how knowledge emerged from activities distributed across time, place and circumstance. Case B is generally regarded as being technologically innovative and has paved the way for application of novel digital tools and technologies into daily fieldwork routines (especially photogrammetry and advanced spatial recording systems). I visited this project for ten days during its 2019 summer fieldwork season, and my visit focused on the challenges that emerge from the use of advanced information systems, including how participants attempt to resolve those problems. My work at Case B was more intensive, and presented me with a fresh alternative to what I observed in Case A, which follows more traditional data management procedures.
- [10] In both cases, the data I collected largely pertains to fieldwork recording practices, processing and analysis of finds, records management, interdisciplinary collaboration, decisions regarding writing and publication of findings, and discussions of how data and findings are presented, evaluated and revised among broader research communities. They each illustrate of the pragmatic and multifaceted ways in which participants reason and work their way through the rather mundane activities that archaeologists commonly undertake in similar research contexts.
- [11] Commercial archaeology is absent from this study's scope, despite the fact that it accounts for the vast majority

of archaeological work conducted throughout North America and Europe. My own lack of knowledge about commercial archaeology contributed to my decision to exclude it from this study. However, Thorpe (2012) and Zorzin (2015) have done similar work in this domain and have arrived at complementary conclusions.

Data Collection

- [12] I assembled a rich and heterogeneous dataset comprising recorded observations of research practices, embedded interviews with archaeologists while they worked, retrospective interviews in non-work settings, and documents produced and used by project participants.
- [13] Observations were usually video-recorded and documented what people actually do, which may differ from what they say or think they do. The main focus of observations was on how people used information tools and interfaces, to account for how the specific context of an activity influences its implementation, and to document how researchers use unconventional solutions or “hacks” to solve problems (Goodwin 2010). Embedded interviews were conversational inquiries conducted while participants were actively working. This helps me understand how and why people make decisions in real-time, considering the immediate pressures and circumstances of their work. Unlike simply observing their actions, these interviews captured the practical perspectives held by the researchers (Flick 1997). Retrospective interviews were longer discussions held outside participants’ work environments. They provided greater understanding of the broader contexts of work by exploring aspects that were not easily observed, such as project planning, publishing, and collaboration (Fontana and Frey 2000). They also provided insight into how researchers saw themselves and their roles within the larger academic community. I examined documents including recording sheets, photographs, labels, databases, datasets and reports to gain insight into the norms or expectations concerning the presentation of legitimate forms of knowledge. I emphasized how people interacted with these media and ascribed value to them.
- [14] I also wrote extensive field notes that remarked on key moments from observations and interview sessions, descriptions of unrecorded activities and conversations, and reflexive journal entries.

Data Analysis

- [15] Through abductive qualitative analysis of observations of archaeological practice, interviews with archaeologists as they worked, and of documents they created, I captured diverse experiences and outlooks that provide insight regarding how data management infrastructures are valued and used in a variety of contexts. This involved coding segments of interview transcripts and observational records and performing systematic writing and conceptual modelling exercises directly alongside the data (Charmaz 2014). This allowed me to build theories based on empirical evidence that reflected the informants’ diverse experiences.
- [16] By listening to participants’ views concerning the systems with which they engaged and explanations as to why they acted in the ways that they did, I uncovered some underlying assumptions that shaped their perspectives and actions. Specifically, I was able to articulate systemic factors that scaffolded participants’ actions, attitudes, and experiences as they contributed to collective operations.

Findings

- [17] Analysis of archaeological materials and data is dependent on systems of control. To control the flow of materials and the structure of data, it is necessary to control people’s behaviours. In other words, the

epistemic value that a project produces is inherently dependent on the control that may be imposed on archaeologists' behaviours. Here I document some ways in which projects scaffold archaeologists' behaviours through technical and administrative means.

Finds processing

[18] Both cases adhered to a relatively common hierarchical organization structure in fieldwork settings (Figure 1). The project director designated a field director who oversaw all fieldwork. A series of trench supervisors, who were usually graduate students or experienced fieldworkers, led and coordinated data collection in specific trenches or areas of a site. Trench assistants, who were usually less experienced students or local labourers, operated under the guidance of their assigned supervisors. Some specialists, including conservators, photographers, and those responsible for collecting spatial data, also worked in field settings.

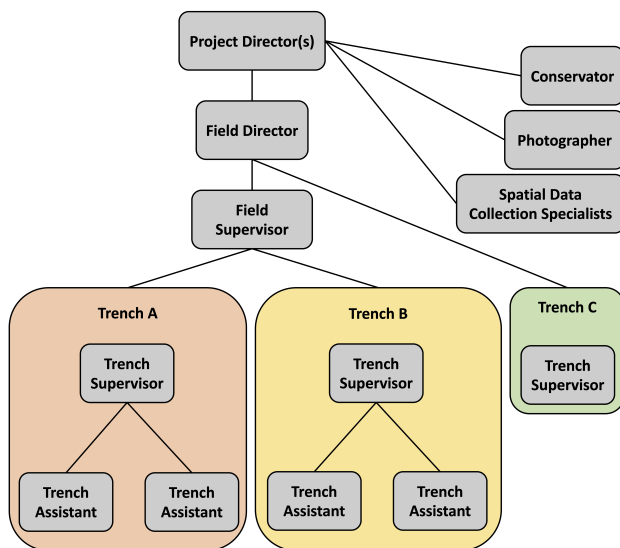


Figure 1: Organization chart pertaining to archaeological fieldwork.

[19] The projects that served as cases uncovered a lot of material, which had to be sorted and processed to facilitate subsequent analysis. Finds processing began in the field, where objects were first recovered and identified as meaningful sources of archaeological insight. At both cases, fieldworkers separated objects into bags and buckets labelled with the excavation unit in which they were found and according to their material properties (i.e., ceramic, bone and shell) (Figure 2). Trench supervisors were almost always responsible for writing on or otherwise dictating the tags and bags, which systematized the bucketing process according to a concrete and lasting official schema. Supervisors were also responsible for instructing assistants regarding how to process materials and separate them out.

[20] Additionally, fieldworkers tossed away stones or debris that exhibited no anthropogenic qualities. They often used sieves to separate loose, non-anthropogenic sediment from small objects, collecting things that they deemed culturally meaningful while discarding the rest (see Figure 3). Sometimes fieldworkers set aside materials that they were unsure about, while documenting the findspot in more detail — “just in case” — before proceeding with their work.^{A1} This documentation was tentative, but may have become more permanent if a specialist later determined that those finds were worth saving for analysis.^{A1,A2}

[21] At the end of each day in the field, fieldworkers hauled all the finds back to the dig house so they could be cleaned.^{A3,B1} Trench assistants spent the next hour or two scrubbing the finds with water, remaining



Figure 2: Sorting archaeological objects into bags based on their physical material properties.



Figure 3: Preliminary sorting of archaeological materials in the field. **A:** Placing unsorted material into the sieve; **B:** Using the sieve to separate loose sediment from larger objects; **C:** Sorting through the larger objects, and keeping the anthropogenic objects in the bucket nearby; **D:** Dumping the non-anthropogenic material into the spoil heap; **E:** Creating a bag and a tag, which indicates the kind of material contained in the bag (“lithics” is shorthand for chipped stone made from chert in this case), as well as information about the excavation unit and context of discovery (i.e. date, excavators’ initials); **F:** Placing the artefacts in the bag.

productive even during the late afternoons when it was too hot to perform strenuous outdoor labour.^{B2} Meanwhile, trench supervisors would do the “paperwork”, which involved transcribing rough shorthand into proper notes and copying handwritten observations into the database.

[22] In my unrecorded observations of finds cleaning I noticed that people talked about the day’s work and their personal and collective experiences, while speculating about the things they recovered and which they were then revisiting in a secondary context. They sometimes even remembered specific finds out of hundreds from the same excavation unit. If they were working in a sensitive or especially interesting excavation unit some fieldworkers would spend a bit more time examining the material that they recovered themselves. Even though they were non-experts and their relatively amateurish analyses remained unexplored, fieldworkers still felt an impetus to examine these objects and try to develop insights from these experiences.

[23] No aspect of finds processing involved maintaining a journal or recording substantial interpretations of the materials, which revealed a failure to recognize that any meaningful insights could arise during these stages of work. It is also noteworthy that fieldworkers were expected to classify finds on the basis of characteristics commonly recognized as more natural, while study of their anthropogenic qualities was typically reserved for finds analysis.^{A4} In effect, a distinction was made between fieldwork, which deals with *materials*, and finds analysis, which deals with *artefacts and ecofacts*.

[24] Moreover, even when specialists worked in the field, they operated as fieldworkers; their actions were constrained by the practical circumstances of fieldwork and the position of fieldwork within a broader apparatus of archaeological knowledge production. In this sense, the division of labour was partly due to pragmatic concerns, namely the fact that fieldworkers were usually occupied with their own specific tasks that required mental concentration, especially when instructed to do this work quickly.

[25] Finds processing was therefore positioned as a means to an end — the end being the arrival of cleaned and sorted finds upon the specialists’ desks, where meaningful archaeological knowledge would be produced.^{A3,A5,A6,B1} Trench assistants were encouraged to act as mere sensing devices, as tools exhibiting an air of objectivity.² This distribution of labour ensured that the recovered materials were properly embedded within a broader site-wide system of archaeological knowledge production, which trench assistants were not (yet) fully aware of.

Finds analysis

[26] Similar tendencies were observed as materials shifted to contexts of finds analysis. After cleaned materials were left out to dry in the sun, they were brought to a space dedicated for finds analysis and storage — usually a museum annex or workspace.^{B3} The dried materials were dumped onto a broad table, and specialist’s assistants quickly sorted through them. They discarded any remaining non-anthropogenic materials collected in error, and counted and weighed all remaining artefacts. Then, more in-depth analysis could be performed based on the finds’ typological and technological characteristics.^{A7}

[27] At Case A, I observed how Jolene, a lithics specialist, went through and sorted the material on the table, sometimes mumbling her reasoning aloud (Figure 4).^{A8} She physically grouped objects, following a general pattern for each excavation unit: cores at the top left corner, blades at the top right, and the extremely high volume of flakes occupying the main space in front of her. After she settled on her arrangement, she dictated her description of the assemblage as a whole, in a fragmented tone that insinuated strong intention behind each word, and Maude, her assistant, wrote down this speech.^{A8} Maude then recorded the quantities

²See Batist (2025a), where this is addressed in greater depth.

of artefacts in each category, as laid out by Jolene, and entered this information into a spreadsheet.^{A8} Sometimes, Jolene would create a new category on the fly, and Maude would push back by asking whether they should slot it under an existing label; those conversations usually ran on for a while, addressing the broader history of different classification schemes. Maude also maintained a checklist to track all aspects of work performed on the materials, including information about the materials' provenance, who processed the materials, when the work occurred, and detailed parameters that structured processing tasks.^{A9} This checklist resembled the records maintained by the project's database manager to track progress and verify the integrity of work.^{A10,A11} In effect, Jolene was primarily responsible for creative knowledge production and for contextualizing the work in relation to a broader community of practice, whereas Maude was responsible for pragmatically embedding these insights within the project's organizational apparatus.^{A12}



Figure 4: Finds analysis in the museum workspace. Jolene (front left) examines the material spread out on the table, while Maude (back left) records her observations in a spreadsheet. The author (back right) takes observational notes, and the project director (front right) does independent work to the side.

[28] A tension is evident here between the allure of concrete representations and disciplinary creativity, which parallels the distribution of creative labour in contexts of finds processing. However in this situation, the assistant was the one responsible for using formal protocols to reign in her supervisor. The power dynamic is therefore not as clear-cut as a hierarchical chain of command, and instead represents a discrepancy between the relatively wild landscapes of domain knowledge and the boundaries imposed by formal data structures.

Integrating and aligning perspectives

[29] After performing their work, specialists reported back to the project with their findings. This involved providing the project with spreadsheets and typeset documents, and sometimes also lab journals, paper

recording sheets, source code, and digital images.

- [30] Findings recorded in spreadsheets comprised formally organized results of systematic analysis. They were organized on a sample-by-sample basis, associated with discrete units from which each sample element was taken.^{A13,A14} Specialist reports, on the other hand, were typeset documents containing summaries of specialists' findings. They described overall trends that specialists observed in their preliminary analysis of the findings, highlighted their potential significance, and proposed ways to extend the work or relate the findings to other aspects of the project.^{B4} As such, specialist reports supplemented the spreadsheets by contextualizing the data presented in the latter.
- [31] Specialists' spreadsheets were integrated into projects' relational databases by importing the relevant records as their own sets of tables, with explicit links drawn between them and the database backbone, which reflected the organizational structure of archaeological entities encountered in the field.^{A14} For example, samples that were drawn from a particular excavation unit were typically listed in a specialist table, but were linked to an overarching index of all excavation units.^{A13} Analytical results could therefore be retrieved for particular parts of a site, alongside findings derived from different kinds of analysis performed on material deriving from the same contexts. The material that was physically distributed and sent out for specialized analysis was thus conceptually re-integrated in informational terms. Diverse kinds of material were rarely examined in close physical proximity outside of fieldwork, though the potential to integrate abstract measurements about them was highly valued.^{B5} However, this actually rarely happened in ways that leveraged the computational potential of these efforts. Specialists provided their own preliminary analyses in written and oral reports, in which they examined the materials under their own purview.^{B4,B6} Analysis was therefore siloed, and findings deriving from these analyses were subsequently integrated when drafting reports or preparing journal articles.^{A15,A16,B7}
- [32] While specialists' interests may have overlapped with those of other members of the project, directors preferred to have one specialist for each means of investigation.^{A17} At the same time, some degree of overlap among specialists who examine the same kinds of materials from different methodological perspectives was expected and encouraged.^{B8,B9} For example, the geoarchaeologists at Case A shared a strong affinity based on their mutual work with geological materials, despite the fact that they specialized in different geological processes.^{A18,A19} This benefited the project immensely, since it allowed each specialist to reflect upon the limitations of their approach and to harmonize that approach with those of the other specialists. However, the value of these complementary perspectives has yet to be demonstrated when presenting findings to the broader research community. Much of the discussion among geoarchaeologists at Case A was process-oriented, including consideration of where to excavate, where and how to take samples, and the degree of confidence that should be placed in certain findings. This information was not effectively represented in the database and in reports, which were more meant to articulate research outputs in a clear and confident manner. Inputting analytical findings into a database was therefore a lossy process that failed to capture much of the pragmatic knowledge and critical reflection provided by individuals and collectives.

Informal consultation

- [33] As the project moved forward, I observed how Case A leveraged consultants' authority on a subject to help contextualize the findings during dissemination.^{A20,A21,A22} The archaeological site that formed the basis for Case A was extremely colluvial, which complicated any effort to date stratigraphic units. This also contributed to many artefacts having extremely weathered surfaces, which made them more difficult to characterize based on their physical properties, though an experienced analyst could manage. Additionally, the extreme amount

of weathering made it very difficult to photographically capture artefacts' physical features, which made it difficult to convince others regarding the validity of specialists' identifications. Sharing illustrations was also deemed inappropriate because of a perception that they were too interpretive and lacked objective legitimacy.

[34] Despite his confidence in his project's claims, and his acknowledgement that the stratigraphy was imperfect, Basil, the project director, knew that he would have a hard time convincing others of the nature of the things that they found. He therefore invited Denise, a major figure in lithics analysis who was responsible for publishing the most definitive assemblages of the same kind of materials, to provide her informal perspective. The rationale behind this decision is articulated in Figure 5:

Basil: ... for decades, there have been accepted means of scientifically representing the archaeological record, umm to act as evidential bases for your claims. It's understood, that you know, it's impractical for every single interested archaeologist to come to see your site, to see your material. You could be in Australia and the site could be in Peru, umm and so, without that tactile, immediate relationship with the evidence basis that somebody's uhh excavated, there are ways of representing it through photography and line illustration. Umm, that always used to seem to be enough, but it seems like again, it comes down to this contentious nature of our site. The fact that we are, we allegedly have such an early site where such early sites are not meant to be, I'm starting to discover that some of what used to be the accepted means of, accepted ehh evidential umm bases, don't seem to be good enough for everyone. And so it's been very important to have people like Denise come and see this stuff in the flesh. Yes, she's seen the drawing, yes, she's seen the photograph, but for her to see it first hand, handle it, pick it up, query it, umm look at, look at assemblages, as opposed to cherry picked illustrated pieces, has been fundamental to convince her that we really have what we say we have.

Figure 5: Basil explains his rationale for inviting Denise to examine the archaeological material.

[35] While Denise was happy to lend her expert eye, she also informed Basil about some of her concerns with the material. In Figure 6, Basil recalled how Denise's constructive criticism provided him with an understanding of how to proceed to accommodate these issues.^{A23,A24}

Basil: So the superstar material that Lauren was excavating, that we thought was Levallois blade cores, she disagreed. She thinks the material is fabulous, she's never seen this material before in Greece, she thinks it's Aurignacian and looks much more like what you would find in France. She's blown away. She's blown away by—
Zack: But it's still Middle Pal?
Basil: Very early Middle Pal. But, we had shown her our superstar pieces. This is where it—
Zack: I was in the trench that first day when we, and I was like cores, wow, tons of them.
Basil: I mean that stuff is— but we had shown her some of our superstar pieces from the survey, like here's a Levallois point, classic thing. But she's like hmm, it's just one piece. Haha you know, it's just like, she's not exactly [unclear], but she has very high standards.
Zack: So is she sticking with, is she saying that Middle Pal was a thing though?
Basil: She was like, so she was looking at these, these little bits and pieces, and like, well, the stuff, you know, we showed her a whole bunch of stuff, it's like, we think this is Levallois, she's like no, it's this. And then we show her some stuff from the survey and it's like, hmm it's one piece and it could be, but it's just one piece, that's not really enough. And then finally we showed her the material, Jolene showed her material from [the trench], Alice's new trench, and she's like yes, this is Middle Palaeolithic.

Figure 6: Basil explains the feedback he received from Denise, and what this means for future work.

[36] From these examples we see that members of the archaeological community who were not actively involved in the project sometimes supplemented the input provided by project specialists. As people who were external to the project, they were not committed to following through on their advice; they instead relayed their

interpretations to specialists who were already members of the project and who were capable of implementing their guidance (Figure 7).^{A25,A26} Moreover, as outsiders, they were not forced to adhere to the systematic recording systems that scaffolded internal procedures, and the informal nature of their contributions was in fact a crucial aspect that made their contributions valuable.

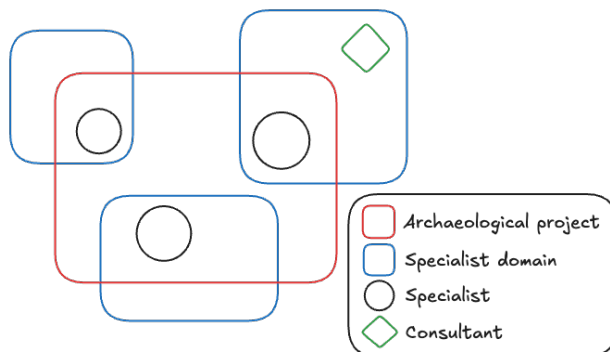


Figure 7: Diagram depicting the relationships between domain specialists who work for an archaeological project and consultants who are brought in to advise. Archaeological projects host specialists from various intersecting domains, who consult with other specialists in their areas of expertise who are not members of the project.

Discussion

[37] Throughout all of these observations there is a clear tension between the need to present information using formal data structures and their failure to fully capture archaeological understanding. On one end, archaeologists are pressed to fulfill their scientific mandate by ensuring that the outcomes of their effort are and appears to be decisive, consistent and stable — qualities that instill confidence in the evidentiary character of data. At the same time, archaeologists struggle to make their data fit a standard mold. This tension produces an “epistemic anxiety” whereby archaeologists recognize — and fail to reconcile — discrepancies between their commitments to formal scientific standards and their intuitive understanding of the limits of this approach to knowledge-building (Huggett 2022).

[38] To escape this anxiety, archaeologists have established workflows that offload decision-making to people who are not immediately present at the site of archaeological encounters (Batist 2025b, 2025a). By re-framing archaeological encounters as a series of programmatic operations and decision-trees, database managers and project managers claim creative agency over how to represent archaeological objects. This removes the burden of decision from the moment of the archaeological encounter, and re-casts those who record what they see, feel, smell and hear as devices that simply match their sensory experience to predefined criteria provided to them. Put another way, workflows downplay interpretive potential at the trowel’s edge by forcing adherence to a series of pre-formulated decisions and by presenting anything other than the outcomes of those decisions as unnecessary addenda to the true or official archaeological record (Batist 2024).

[39] However, while workflows are ways of coping with the epistemic anxiety, the tension that produces it is an ever-present aspect of archaeological research.³ In fact, workflows are constantly being tested and resisted. Archaeological objects and phenomena are “wild” in the sense that they were not made for the benefit of archaeological understanding. The eclectic, improvised, opportunistic and nimble nature of archaeological

³And of all scientific research.

knowledge production is an adaptation to this fact — born out of necessity to respond to the slippery and inconsistent character of archaeological encounters. The inability to fully control archaeological practice is therefore not due to intentional acts of rebellion, but an inevitable outcome of the myriad ways in which archaeologists do their proper research.

- [40] This paper articulated some specific examples in which these persistent challenges were manifested, drawing special attention to relationships between people occupying different roles and the formal data management systems that governed their work. Specifically, I noted how members of archaeological projects were recruited to either represent the interests of the database or of the things to be recorded by it. For example, in the situation of finds processing, trench supervisors were responsible for ensuring that the materials were channelled into discrete buckets, in accordance with the broader pipeline in which excavation is situated. Then, in contexts of finds analysis, I noted how a lithics specialist relied on her extensive training to systematically classify the materials, while her assistant served as the database’s enforcer. Subsequently, the specialists’ relationship to the database changes after calling in the consultant; the primary lithics specialist was free to update the data model with new considerations deriving from the consultation, but had to re-formalize them into a new authoritative data structure.
- [41] Despite these challenges and tensions, archaeologists were still able to generate meaningful insights about the past. They accomplished this by developing ways to easily navigate between tight and loose control; on the one hand, confident representations of archaeological encounters are deemed especially valuable for producing scientific outputs, but this was tempered by a common understanding that that approach is a limited window upon which they may view the past. This flexibility to think of both the potential productive outcomes and limitations of different forms of engagement enabled them to understand when it was possible to bend the rules, when they needed to account for context, and when they needed to be more structured.
- [42] Moreover, the capability to actually enact this flexibility derives from social and communications strategies, rather than purely technical solutions. This understanding is embedded in archaeologists social and collaborative relations, rather than in any technical apparatus.⁴

Conclusion

- [43] Practically speaking, archaeological records are created through human intervention, whereby people ascribe meanings to materials, delineate natural phenomena from cultural processes, distinguish signs of modern intervention from activities performed by the archaeological other, and select signal from noise. And as with all things of anthropogenic origin, the records that archaeological produce carry with them traces of the extended social and material worlds that scaffold the communities that produced them. This paper articulates some collaborative and organizational features of formal archaeological data management systems that have such a large impact on the outcomes of our collective efforts.
- [44] Specifically, it documents how systems designed to direct the flow of information do so via the coordination of labour, and the strategic arrangement of human and object agency. Of course there is great potential value in having access to centralized and tidy data deriving from tightly-controlled workflows. However, workflows should be recognized as *attempts* at making data fully transmissible and interchangeable, which are resisted by the slippery nature of the things we attempt to document and the myriad ways in which archaeologists

⁴It is notable that data scientists prefer to work with little understanding of how data came to be. Their power derives from their ability to take data at face value, and for that to be considered a valuable quality. Archaeologists also have people who deal with data, but we call them archaeologists since they are part of our community of practice and have obtained this archaeological sensibility through fieldwork and other embodied experiences.

attempt to capture them. These gaps, frictions and sources of resistance need to be more fully accounted for in philosophies that underpin data management infrastructures and policy frameworks — which are ultimately mechanisms of control — in order to ensure that these services actually serve archaeological needs, values, expectations and priorities.

Supplementary Materials

[45] This supplement contains sections of transcribed interviews referenced throughout the paper. References are ordered by case and by the order in which they appear. Each reference has a prefix corresponding with the case to which it pertains.

Case A

A1

[46] Without cleaning, and perhaps just to see whether it is worth her time, or to check if the deepest spot she reached is adequate, she measures another elevation; she can then go down to that point knowing that it works.

Actually, it was to measure the depth of a large handaxe-looking thing. She noted it earlier but I guess she waited till her legs got sore to take an elevation. I may ask later.

She did not use the term x find. She is unfamiliar with handaxes, [illegible] particular a preform, but apparently Basil got excited about one on the other side yesterday, and because she's a newbie she decided to record it just in case.

A few minutes later Olivia uncovered a large phallus shaped stone, similar in size to the large handaxe, and we all joked around about it.

Nina at the sieve, brought over an orangey-brown piece of chert, no cortex, for Olivia to examine more closely. "If it weren't this colour, it might be shitty material. I would chuck it. But so [illegible] heat treatment caused darkening colour so this might be that."

A2

[47] At around 7:55 Will asks me to put a little nodule in the soil sample bag. I may ask him about that later.

A3

[48] **Jolene:** Yeah because you can't study dirty lithics

Agatha: I mean that's also connected with the schedule, we changed the schedule of the new place we worked and we are supposed to stay there until five

Zack: What do you mean by the dirty lithics?

Agatha: Well the lithics came dirty again

Jolene: They're not washing them properly, so how can you make something

Zack: Do they get better as time goes on?

Agatha: Yeah, but I mean the thing is like no one is thorough enough to empty the bag from the soil, and then put the washed lithics inside, and no one is there to actually control the, because we have a new schedule so I mean we put some people in charge but...

Zack: Pardon my sort of half-feigned ignorance for the sake of the interview, what does dirty lithics, what are the problems that they cause?

Jolene: Well you can't study them

Agatha: You can't see anything

Jolene: They bring them to the lab and you're supposed to look at them as say this is you know, i dunno

Agatha: Sometimes even when they're washed properly and the raw material is a bit like, we're not used to that raw material, it's hard to see the ridges and everything, but like even if they're not washed properly then that's another headache.

Jolene: It stops us from doing our, so we need to do their work

Agatha: The worst was like talking with Denise and trying to pursue like what is re-touched and what's not, and Jolene was like pushing one small piece saying like this is retouched, and they were like no it's not, and then I just went and wash it and there was a completely fresh, it was white haha and it was retouched I think, it sucks.

Zack: But did she think it was retouched?

Jolene: Yeah

Agatha: Yeah

A4

[49] Jolene separating diagnostics from non-diagnostics. Non-diagnostics get pulled aside to Agatha's side of the table, where she separates those into ecofacts/natural/too weathered to matter. She throws those in a zimble, and counts them off. This is counted, and subtracted from the number recorded the other day, in error, which included both natural and cultural together.

A5

[50] **Zack:** Are you including any supportive documentation, such as data tables, figures, details, methodological descriptions or metadata?

Basil: Yes. So umm if it's finally accepted, it will be in the journal Science Advances, which has umm, it was, in the end it was about an 8000 word limit, uhh which I think, it's fairly short. I mean originally when we wrote it, we wrote it for Nature, umm which was like a 1500 word article. Umm so... which is like very concise. Umm, now, with all of those venues, you know, it's more and more common that people obviously are reading stuff online, all of these venues, including Science Advances, should they ultimately publish it, umm we, they facilitate and encourage the publication of supplementary online material. So to an extent that's where you can give a little, you know, a certain amount of data dumping. Umm, where you can go into the really nerdy— and in a way it's great in terms of like, you know, your main article, it's a relatively flowing piece of narrative. And then, you know, see supplementary online materials for a detailed breakdown of the stratigraphy, for a more detailed discussion of the lithics, for a more detailed breakdown— actually no, at the end of the paper you're then allowed a materials and methods thing, which is where you go into great detail for the scientific umm uhh audience, uhh the nature of our dating techniques, et cetera. So, quite snappy archaeological big picture narrative, expected materials and methods for the, sort of, the main paper, and then supplementary online materials, that's where you can go, you know, you can write hundreds of pages. Umm, there is an expectation... so, this is an open access journal, umm.

Zack: Entirely?

Basil: Yes.

Zack: Not a hybrid or...

Basil: The Science Advances one is open access, which if we're accepted will cost us 4500 dollars US, and one of the things they ask us... so, each author is expected to sign a quite detailed form that I've not hitherto

dealt with before, which basically gets you to stipulate quite clearly what your contribution was in the paper. So at the end of the paper is says, you know, use your initials, [redacted] for myself, umm director of the project, I kinds worked on the lithics, co-wrote the paper, and then you know, other people, geoarchaeology, blah blah blah, and et cetera. But then we also, online, have to sign these documents, in terms of, you know, did you produce any algorithms? Yes, no. Umm did you, were you the person who raised the main funds? All these projects, in great detail. Umm.

Zack: What do you think the value of that is?

Basil: Umm, given that it's... I'm not entirely sure why we had to do it. I'm sure there was some kind of blurb, which, you know, it's one of these things that you have to say oh I accept, you have to do it, as opposed to really ponder it. Umm, that I don't think is published, so whether there's, who evaluates that, I'm not quite sure, or it's exact purpose. I mean there is a slight discomfort, potentially, in that one of the authors is essentially a political appointee that we know has to be included, umm and I'm not quite sure where on that long yes/no list they would have hit yes. I don't know if that would then trip anything from the journal in terms of like he can't be involved, you have to have written something or given something. Umm, but yeah. So, but umm one of the other things they said, is you have to be crystal clear in terms of you have to provide your primary data that supports your arguments. Or, and we took the or route, because we're still organizing our data to be in a umm, a more accessible format, umm or basically you say it will be made available should you contact us. So we've, we've included that required statement. If anybody wants, you know the uhh, the numerical data that underpins any of this stuff then we will happily hand it over.

A6

[51] **Zack:** Umm, have you worked in other settings? Like you've worked in the museum, right? Just lifting boxes, right?

Jane: Yeah. But...

Zack: Yeah. And you've just been in the field I guess, right?

Ben: I've been in the museum twice. I've lifted boxes like Jane. I've also done the counting and a bit of lithics washing in the museum.

Zack: What did you think of it?

Ben: Umm, you wanna go first?

Jane: Not my fave, to be honest.

Zack: Yeah.

Jane: I just, like, I don't... maybe if I knew more about what I was looking at then I would be more interested. Like I'm sure if it was like a bone lab I would love it. But it just like, lithics are cool but they're not really my jam and I don't really know what I'm talking about, so like, working with them is kind of like, it's like counting rocks. You're required to count like scapulas and unclear

Ben: Yeah. I was basically the same way. I found it interesting at first for a little bit, and then it just became monotonous and like boring and just mind-numbing, basically. And the thing you said about, just, you don't have the expertise in it, I felt the same way. Like I was just looking at lithics or rocks that I would have picked out, and I can't say like, oh this one is definitely not. So I'd be like, oh, Agatha, is this something?

Zack: I actually haven't taken a lithics class yet, or at all.

Ben: Well I'm planning to next semester, next year, but I had to—

Zack: Was it because of this project, or no?

Ben: Yeah, basically. No no, it was, for sure. It was because Basil, and I have some experience with it so I'm gonna do better, hopefully.

Jane: Hahahaha

Ben: In theory.

A7

[52] Moved to watch Agatha now. She pulls a pile from the larger pile made by Jolene. Separates cultural material from natural material to throw away. Cultural material is out in a pile in the back, natural is kept where it is, directly in front of her. Once the pile in front of her has no cultural material in it anymore, she separates the natural to the side and clicks her clicker according to how many are moved. Then she sweeps these off the table, into the zimble on the floor. Or, to be quicker, she clicks as she drags them into the zimble, or as she drops them in after grabbing a handful of small material. This number will then be subtracted from the total count, which was mistakenly recorded earlier without removing the natural materials.

A8

[53] Jolene describing laid out materials. Somewhat of a performance. Starts with a big core from Lauren's trench. It stands alone. Tosses is around, across her hands, but looks at Agatha, who takes dictated notes, as she speaks. Speaks a few words at a time, [illegible] so that each four words are enunciated for Agatha. Concludes by "and that's it" before adding a last minute additional note. Then adds "and that's enough". Onto [redacted context ID], laid out already. Looks down at the table, scans the material. Does not touch yet. Generalized descriptions. Most, many, boolean logic, etc. Describes a few particular objects in more details. Hones down into more specific groups of finds, describes tendencies among them. Still very descriptive. Stating how many of what. Defining the scope of the assemblage. Not yet uttered anything about periods, etc [illegible]. After describing patina, however, she begins to add some interpretations of deposition conditions. Then notes burned pieces. Now explicitly notes "concerning dating of material..." Notes caveat first, conditional clause proceeding what her observations are pertaining to a subset of the material, very cautious and selective with her words, and does not backtrack or correct herself. Then explicitly introduces her notes on the emery. Then moves to the side, has closer look at the smaller finds. Physically handles a few. Lydia asks to put them back in the bag. Jolene says sure, but goes through them, selecting a few to bag separately, for [illegible] to go through later for a second, more specific look. Lydia's question to pack up was a welcomed interruption. Jolene was a bit fed up with this context, since it has been on the table for a very long time and she was glad to see it settled. Agatha writes down what Jolene dictates in a small notebook. She will then transfer these descriptive notes to the database in a more standardized and "nicer" language. Now for [redacted context ID]. Starts with material selection. Most to least. Notes mixed material, based on partial and "[illegible] typological [illegible]", does not specify yet. Now describes Mesolithic. Describes most abundant [illegible] of lithics. "As well as one multidirectional core", it stands out as exceptional. "Second group would be Upper Pal tools and blades in similar preservation". Then goes on to describe subgroups within it. "This assemblage is defined by..." Then moves on to Lower Pal, as well. Grouping emphasizes patination, since this is important for discerning Lower Pal material.

A9

[54] Agatha does a lot of busy work. She brings material back and forth from storage, for Iona to count and weigh, and for Ana to analyze. She also prepares level 1 recording sheets where counts/weights can be written down officially. Also compiles the list of already closed contexts, numbers of bags within, etc. Planning work for [illegible].

A10

[55] **Basil:** We're still, I always sort of gave, because Jolene's been here from the get-go she has sort of first dibs on what sort of material she wants to work on, and I think she was always keen to work on sort of Middle to Lower, but until this year we didn't really have much of that. Now that's crystallizing a bit more for her, whereby it makes sense for me to bring in somebody else to work with me on the Upper Palaeolithic or Mesolithic.

A11

[56] It is extremely difficult to share how good I feel about this code. It represents a major breakthrough in my learning how to use R, and it sucks that no tangible research products can come out of this in a way that shows my skills. I actually included an R script on my [funding](#) application that was due on Friday, as a non-refereed contribution, but that designation is sketchy.

This seems to be true of database work in general. I will likely not be published on major papers as a co-author because my work is 'supportive' but not necessarily "creative". The output of database work is other people's work. But the product, the database itself, is a creative output. It's the result of a tremendous amount of unpaid work, highly skilled work at that. However, there are no ways to recognize or evaluate these skills. At least not in a formal or bureaucratic way. Perhaps this is why there is so much gatekeeping and nerd-signaling among "digital archaeologists", it's a way to indicate that one's in the know. But I don't believe that that is a reliable indicator of skill or know-how, it's easy to play the game on twitter, it takes lots of time, over the long term, to develop digital skills.

A12

[57] She actually really hates these organizational tasks. She is really grateful for having Agatha as her assistant, who could do all of that, which she generalizes and dismisses as of somewhat lesser importance, or as the distractions from her more hands on work.

A13

[58] **Zack:** Uhh in your view, what are the main challenges or considerations in managing information productively for archaeological... can you just mentioned an aspect of that?

Basil: Sorry, managing...?

Zack: For managing information.

Basil: Umm

Zack: In archaeological, which may involve fieldwork or other aspects of work.

Basil: Umm, it's, part of it is actually sort of umm, a combination of trust and communication, and occasionally a bit of coercion. There's, my particular role is to like, you know, you let certain people, you're bringing people who you believe, or know, can do their job, and that job is often the generation of data that's pertinent to answering your research questions. Umm, and different people work in different ways. Some people are very gregarious, other people are like hyper paranoid about keeping stuff, and you just gotta, you

know, as long as there are deadlines and you need communicate how you're gonna need the data, which again, that's a steep learning curve, often at times you just assume that by osmosis somebody is gonna know what you're gonna want by the end. Umm, and you know. And now I'm coming to appreciate that, you know, there's a greater degree of me being hands on and centralized, and it's like guys, everybody has to work in this particular system. You know, your end products will need to be coded and accessible in this particular way. How do you get to generating those data? That's your skillset, how you work, fast, slow, you know, et cetera. That's up to you. But ultimately we're gonna need an artefact ID system. However you deal with your pottery otherwise, or your ceramics or your ground stone, you are going to need to interact with this point. Umm, and, you know, this is a new project, a number of us are young scholars who— yes, we've all worked on other projects, but you know, more often than not those projects have their own systems, so it's a matter of like, you know, figuring out, you know, what we need. And some of this is probably gonna be question-led. Other parts of it is just like no, this is just, this is what everybody does. And other parts of it might be from external forces. You know, it would be the, the [national] Ministry of Culture requires us to provide an inventory of our crates.

A14

[59] **Basil:** and where you tend to have the umm, I mean there's, in terms of mediation, there's mediation in terms of like, Zack, I need you and Dorothy to get together so that Dorothy's skills and data dovetail and is integrated into the database. So that's like okay, I recognize an issue, lets make the space for this to happen, and try and mediate and then okay I think you guys are bright enough to talk to each other, and then you can come back to me in terms of like do you want this or do you want that.

A15

[60] **Zack:** So how do you get ramped up again?

Basil: Umm, usually umm the energy comes externally, or I use external pressures to get me back up and running. It's like, oh, I've got 15 days to submit the next grant application, which with a number of our funding agencies involves writing what we did last year, you know, how did you use your budget, what did you achieve? Umm, so umm, and also the fieldwork committee at the [redacted] Institute. There are, there are two or three sort of umm deadlines coming up, that then kick me back into it, which is going to be a matter of like, okay oomph, now I'm going to have to need to send an email to Alfred, please can you send that photograph, Alfred can you explain what you mean by this, Alfred can you write your report up. You know I got yours ahead of time, I've got your database report. Dorothy has sent through her report, but her report is not very interpretive.

A16

[61] **Zack:** So, moving on maybe to some more uhh, the aspect of data sharing in general, umm how involved were you in the uhh preparation of data that will be shared openly? I mean, obviously it's still in uhh processed, you said that you would have that contact us for uhh... Like, who is it, is it Gabe or, like who is going to be doing that, and how do you anticipate that being, that going on?

Basil: Well the umm, the, there's two main, umm three main sets of data in this. There's the geoarchaeological data, umm which, hopefully we have made accessible. Everything that the public or the readership needs. Umm, there is a slight tempering of all of this in that these data underpin a yet to be completed PhD by Alfred.

Zack: Right.

Basil: Umm. The analysis of the lithics, which is undertaken a level 1 and a level 2, is primarily uhh, if not exclusively undertaken by Jolene and myself. Those data are then fed into a database, which means by extent you and others have immediate access to it. Gabe often then runs with those numbers and does a lot of, sort of, you know, what can we tease out of this patterning-wise, you know. Usually it's a matter of, like, Jolene and I are very much focused on the description of the assemblage, and we generate these data that somebody like Gabe can then sort of interrogate in terms of assemblage structure and correlations, and whatever umm.

Zack: How does that relate to the geoarchaeological? Does it link up in any way?

Basil: Umm, no. Hahaha. So, you know, again, it's, I'm painfully aware of the nature of Alfred's work, in that he's a young scholar who needs a bit more control over his data than somebody like myself. Umm, so, you know, he sent, you know, Alfred's been very good in terms of, he's always been on time or up front in terms sharing data, interpretations, be that photo micrographs, be that maps, be that text, be that quantification, qualification of whatever it is he's doing. Umm, but in terms of the original counts, or all of the various versions of the photo micrographs that he took before he chose version seven, those I don't have to access to, and nor have I asked him to hand it over yet. Umm, because I feel, you know, his rights, in that instance, kind of trump that of the project.

Basil: Umm, so umm, the number crunching that go, you know, the, a whole bunch of the smoke and mirrors that underpins umm the French dating people, you know, we have all the end products, but, you know, the algorithms, the 17 excel files that have been played around with, or whatever it is they do in France, I don't have those. And to be honest, my mind doesn't even work in that fashion, in terms of I need the archive, I need to be able to see every stage in it. Because if they gave it to me it's gonna cluster up my computer and I wouldn't know what to do with it in the first place.

Zack: But what has the publication done with it?

Basil: Umm, like I said, Science Advances requires that we make available, make data available, umm primary data available or make, have a commitment to sharing it should somebody ask for it.

A17

[62] **Basil:** And then there are those more sort of umm conflict management I guess, which is where you have, and usually this tends to correlate with relatively junior scholars, who don't have the security of a job or act more insecure in terms of like their progress, their knowledge and whatever, where it's like we have, I mean basically, you know, to a large extent we're all working, most of us are working with stone tools, and a whole bunch of others are working with soils, so it almost boils down, most of the archaeology, to two components. And yet we have more than two people working on them, which means that we have people who are ideally complementary in the information they're pooling for the project, but I'm also very aware of the fact that while people are inputting to the project, they kind of need their own datasets, there needs to be a bit of a boundary marking in terms of this is ultimately mine, and I will be publishing this. Or at the very least, I will be publishing this as the first author, with this person. So, and I've experienced a lot of that at [a prior project], and so it, for when you have these little flare ups, it's not something that, none of this surprises me. As to how one deals with it, I've never been in that position to actually have to mediate these sort of things. So I've learned, probably, to be more clear up front in terms of this is my expectation of you, what's your expectation, this is yours, this is common, you know—

A18

[63] **Zack:** I noticed one example I found really fascinating was your conversation, your three way conversation with you, Alfred and Andreas, I found that really fascinating as you all seemed to have a different role in the

conversation. I was wondering if you have any strategies that you commonly use, to assess whether they're different, or whether they're integrating or whether they're divergent, or like, how do you sort of assemble these different ideas, and these different knowledge bases?

Basil: I mean, either I initiate meetings or I make sure that, you know, the three people who need to be talking to each other are brought together. I mean what was fantastic last year was having brought in Marie, Marie, Gabe and Alfred spoke the same language, it was wonderful to see. Whereby I was like, after a while, okay, I'm going to go off and talk to these people over here, you guys are obviously perfectly capable on your own.

A19

[64] **Basil:** Well, I think, you know, it, it was great pleasure to you know, sit back across the table and watch, listen to Marie and Gabe and Alfred really talking in great detail about what they thought was going on in the site formation processes, what implications that had for sampling, what implications it had for the interpretative limitations of what they can say, and how things could be bracketed, how things could be rescued. They were already working on the narrative, they were like, it was basically, they were ready to roll with whatever we got. You know, it wasn't a matter of like "this is the result, ooh now what might this mean?", they were like really thinking through what this sort of, where this might actually take us, and they were all, they were using the same language, they really, there was a great deal of mutual comprehension.

A20

[65] **Basil:** It's, I mean, yeah. Umm, now, for a very, you know, for decades, there have been accepted means of scientifically representing the archaeological record, umm to act as evidential bases for your claims. It's understood, that you know, it's impractical for every single interested archaeologist to come to see your site, to see your material. You could be in Australia and the site could be in Peru, umm and so, without that tactile, immediate relationship with the evidence basis that somebody's uhh excavated, there are ways of representing it through photography and line illustration. Umm, that always used to seem to be enough, but it seems like again, it comes down to this contentious nature of our site. The fact that we are, we allegedly have such an early site where such early sites are not meant to be, I'm starting to discover that some of what used to be the accepted means of, accepted ehh evidential umm bases, don't seem to be good enough for everyone. And so it's been very important to have people like Denise come and see this stuff in the flesh. Yes, she's seen the drawing, yes, she's seen the photograph, but for her to see it first hand, handle it, pick it up, query it, umm look at, look at assemblages, as opposed to cherry picked illustrated pieces, has been fundamental to convince her that we really have what we say we have. Ditto Jack. So I think, you know, because I still hear through the grapevine people, you know, muttering, that oh you know, maybe it's this, maybe it's that, you know. And my colleagues over in [the neighbouring region], I mean, what's interesting is that, you know, my colleagues in [that region], Richard and Bruce, they were the first guys to find the Palaeolithic stuff. Denise doesn't believe in that material, so other people do not believe in that material, even though they've done the scientific representation, they drawings and whatever, umm—

Zack: has she handled it?

Basil: I— she's been there, she's been to the site. I don't know if she's handled the material because the material I think was in [another city].

Zack: Do you think that if she were to handle it like she did at [your project] it would, her opinion would change?

Basil: Well I worry that the fact that she's been there and she had a tour of the field, and that's... They

have a problem because they work with quartz. Quartz, having just studied some of the material, umm of Richard's Mesolithic site, it's really difficult. Umm, I just studied the material, and I hope I was consistent in my inconsistencies, but like I worry about some of the site. Was I really seeing what I wanted to see? So umm, the visitation of these scholars... Now, there's scholars and there's scholars. The very fact that, you know, to an extent we just tell everybody else bugger off, we don't need you to come anymore, because we're gonna say "see Denise, pers. comm." Denise is gonna go off and tell other people that what she's seen, she believes in, and that's gonna be good enough as a stamp of approval for a lot of people. So that, that's a different, that's—

A21

[66] **Zack:** Yeah. So back to the work on [this project], how do you see your work contributing to the overall aims, and if you have a conception of what the overall are...

Jolene: A lot hahaha well they're only dealing with all of the material that's excavated and giving our impressions for the reports. So that literally sets up the basics of how we see, the site.

Agatha: There are no bones, there are no shells

Jolene: How we see it as human activity

Zack: There's a lot of pressure, then

Jolene: So, yes. Of course. So ummm there's lots of pressure but it's also very rewarding when we get the confirmation from the outside.

Agatha: Like we did today

Jolene: It hurts when we don't hahahaha so

A22

[67] **Basil:** It's, I mean, yeah. Umm, now, for a very, you know, for decades, there have been accepted means of scientifically representing the archaeological record, umm to act as evidential bases for your claims. It's understood, that you know, it's impractical for every single interested archaeologist to come to see your site, to see your material. You could be in Australia and the site could be in Peru, umm and so, without that tactile, immediate relationship with the evidence basis that somebody's uhh excavated, there are ways of representing it through photography and line illustration. Umm, that always used to seem to be enough, but it seems like again, it comes down to this contentious nature of our site. The fact that we are, we allegedly have such an early site where such early sites are not meant to be, I'm starting to discover that some of what used to be the accepted means of, accepted ehh evidential umm bases, don't seem to be good enough for everyone. And so it's been very important to have people like Denise come and see this stuff in the flesh. Yes, she's seen the drawing, yes, she's seen the photograph, but for her to see it first hand, handle it, pick it up, query it, umm look at, look at assemblages, as opposed to cherry picked illustrated pieces, has been fundamental to convince her that we really have what we say we have. Ditto Jack. So I think, you know, because I still hear through the grapevine people, you know, muttering, that oh you know, maybe it's this, maybe it's that, you know. And my colleagues over in [the neighbouring region], I mean, what's interesting is that, you know, my colleagues in [that region], Richard and Bruce, they were the first guys to find the Palaeolithic stuff. Denise doesn't believe in that material, so other people do not believe in that material, even though they've done the scientific representation, they drawings and whatever, umm—

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A23

[68] **Basil:** So the superstar material that Lauren was excavating, that we thought was Levallois blade cores, she disagreed. She thinks the material is fabulous, she's never seen this material before in [this region], she thinks it's Aurignation and looks much more like what you would find in France. She's blown away. She's blown away by—

Zack: But it's still Middle Pal?

Basil: Very early Middle Pal. But, we had shown her our superstar pieces. This is where it—

Zack: I was in the trench that first day when we, and I was like cores, wow, tons of them.

Basil: I mean that stuff is— but we had shown her some of our superstar pieces from the survey, like here's a Levallois point, classic thing. But she's like hmm, it's just one piece. Haha you know, it's just like, A, she's not exactly ship's been gone, but she has very high standards.

Zack: So is she sticking with, is she saying that M. Pal was a thing though?

Basil: She was like, so she was looking at these, these little bits and pieces, and like, well, the stuff, you know, we showed her a whole bunch of stuff, it's like, we think this is Levallois, she's like no, it's this. And then we show her some stuff from the survey and it's like, hmm it's one piece and it could be, but it's just one piece, that's not really enough. And then finally we showed her the material, Jolene showed her material from [trench], Alice's new trench, and she's like yes, this is Middle Palaeolithic.

Zack: Ah, okay.

A24

[69] **Basil:** Denise came and looked at the stuff and she said: oh, 90% of this material, you should just take it back to the site. And I was like oh, that's a very liberating statement. So, it's like, yeah. Because, because it's a umm, a quarry site, we've just got stupid amounts of stuff. You know, the amount of techno-typologically diagnostic material is relatively low, and particularly with regard to actually finished, modified tools, is very low. So like, you know, 99% of our stuff is like un-diagnostic material, apart from it's diagnostic, but you can say this is Upper Palaeolithic. But like most of this stuff if rolled down a hill, it's mixed and blah and whatever. So yeah, we might be able to say that these flakes are Upper Palaeolithic, but actually the end products span the Aurignation to the Gravettian, so this particular flake, we just, there's nothing you can do with it. Is it worth actually saying that there is a... So I think what we can do, but we have to have this conversation out loud with each other, can we justify, like, so you know, one of the contexts I dug, we had 36 bags this big coming out of one context. Now, there might be a very good argument to make, but we need to make this argument out loud to each other and write it down to see if it makes sense and whether we can

defend it, whereby of those 36 bags, we're just gonna take one bag and record level 2 all of it as a sample, and then we're gonna rapidly go through the other 35 bags to make sure that there isn't a hand axe in it, take anything sexy out of it, in case we want to draw them, and then the contents of those 35 bags, we put it into the car and we take back to the site. And we do that with every single context we've dug.

A25

[70] **Zack:** It's a different kind of engagement than a collaborator though.

Basil: Yes. And uhh, and the rhetoric for me is not going to be so much for investment—

Zack: It's gonna be an arms length—

Basil: Yeah, it's gonna be, in this case, it's like a tactile veracity.

A26

[71] **Basil:** But like in terms of the visitor thing, to go back to that, umm, yes, you, having visitors is partly profile raising for the site, community engagement and community creation and networking, uhh we also brought Denise over in part because we wanted to impress her, but also we wanted her critical input and we were kind of, you know, Jolene and I can't publish all of this. We weren't necessarily looking for her to come in to publish with us, because she's a very strong character and I think she might overwhelm us, but we were sort of hoping to maybe get suggestions from her as to who else we could bring in, say like a postdoc or a—

Case B

B1

[72] **Zack:** hahaha. Umm, so uhh, I guess uhh maybe we'll move onto some questions about, you know, workflows that occur throughout the project. What kinds of, I mean I have to play a little bit ignorant here, so what kinds of material does the project collect? And uhh how are these materials collected? How are they processed and prepared?

Rufus: So just archeo— you're talking about archaeological—

Zack: Yeah, I mean any, I mean I feel like, because I can see that, but I want to see your perspective on that.

Rufus: Sure. So with the excavation we umm approached that in kind of the new world setting in that we have total collection of pottery, uhh total collection of worked lithics, umm total collection—

Greg: Do you need anything? I'm gonna go downstairs really quick.

Rufus: Umm no, thanks.

Greg: Okay.

Rufus: Uhh so total collection of pottery, umm total collection of any archaeological material coming out of the ground, we collect that. Umm starting next year, we are going to do a palaeo-ethno-botanical kind of approach to things. Uhh with that whole endeavour then, it's once we're, which you've experienced here in that middle trench, we've had inside structure materials, and we're working with uhh a colleague of mine, [redacted], umm back at BU, he has his own palaeoethnobotany lab, so over the, uhh over this next couple of months or so when we're finished in the field in preparation for next year, we'll get the sampling strategy umm because we want to have a robust quantitative approach to the collection of those palaeoethnobotanical remains. Umm the flotation and things like that. So total collection of pottery, and we're gonna do the botanical stuff, umm total collection of any other material. Now once that material leaves the field, obviously it's recorded in a very systematic way. As you're aware, it's tied to the stratigraphic unit itself. Anything of note found in situ is also given a special find number, that becomes a special find, and it's inventoried

separately, after the processing is done. Once all the material goes in, ceramics, they're all washed. Umm faunal material is not washed umm. Uhh obviously any metals, we have quite a bit of metals, none of that stuff is washed. The ceramics, once they're washed, that's something that then they are umm analyzed using a umm SUIR form, so stratigraphic unit inventory recording form. What that then does for us is we count every shard, we weigh every shard, and then we batch it up by ceramic ware. Anything within that is that is highly diagnostic or useful umm in the interpretation or dating of the site, we take that stuff to the side and inventory that stuff as well. And so that, then the rest of that material, having been quantified then gets bagged, gets put back in the bag, and that goes into storage. Uhh the inventory stuff, then there's a whole new level of recording when there's an inventory catalogue sheet.

Zack: This is for the diagnostics?

Rufus: Exactly. So the inventory catalogue sheets are then taken and number of, a whole new layer of information is gathered from that. You know, basic weights, thicknesses, umm heights if there are umm, what type of material it is, any type of publications from comparanda that we can bring in. Then that material then becomes the basis for the publication of the catalogue. That, with the generic quantitative information that we pulled from that other material.

Zack: From the non-diagnostic material.

Rufus: From the non-diagnostic, yep.

B2

[73] **Zack:** What's, are there anything that's exclusively Chris?

Rufus: Umm so Chris is umm, so Chris is at [redacted university name] and he's bringing students over this year.

Zack: Right.

Rufus: So he's responsible for that educational component this year. Next year, Logan and I at [redacted university name] will have our own field school as well. Umm so we'll bring out around ten students, and then umm, and at that point Logan will be responsible for that educational component. Chris will maintain his control of his educational for [redacted university name] students. There will be a lot of overlap [unclear]. But umm—

Zack: He's gonna be expanding the project, I guess.

Rufus: Yeah, a little bit bigger. And we'll wander around the same trenches, around two or three, but then having delegated groups for pottery washing, switching out that with excavation.

B3

[74] In the warehouse now, with Liz, Vanessa, Chris and Lisa. Liz, Vanessa and Chris are organizing the washed ceramics, trying to make other work easier later on. Also, by bagging the pottery that is sitting out on trays, the trays can be put to use by pottery washers. They seem to also be going through a first pass of the material, noting things that stick out from the rest, and also re-bagging stray shell or other kinds of finds that do not belong with the pottery.

B4

[75] **Zack:** Do you think that specialists might need some different kinds of, like do they be putting in requests for specific things?

Rufus: So our, no, our faunal person never asked for stuff, but with the the botany stuff—

Zack: Because they provide their own forms, and the templates or whatever, you don't have to...

Rufus: Yeah. They give us a full report, and we then take that report and rework it as part of our department of antiquities report and as part of our publications.

B5

[76] **Greg:** So here I have different tables, and I'm normalizing the database. So instead of just creating one table with all the data, I'm creating multiple tables and developing umm foreign keys and relationships, and the primary keys and things like that. So the umm tables I created is area, umm just umm, and I did that because even though it's [the site], we can use umm the area, if he decides to have another site, we can just use the same database. And it can easily add just another another to there. So this isn't umm order but the ones that we use, so the stratigraphic unit holds all the metadata from the forms, so this form here, it's going to hold most all of this except for features and photographs and drawings. So it's going to be a separate table, because we have many, like, one to many. So it's easier, that's what normalizing the database is. It's easier to create this in a separate table, and then it link it to the main table. So then you don't have repeated data, you don't have to keep typing the same stuff in, umm that's, so that's why we're doing the, breaking those out. But most of this stuff, and finds is also going to be another one, because we'll have multiple umm things, and keeping it separate is easier to query that table and link it to whatever you need to. So we'll go back here. Umm so there's the [unclear] and the photographs and drawings tables that go along with the stratigraphic unit table. And the finds table right there, number of bags. So the next one's ceramics, which is going to be this one right here, so it's going to have this data like in one to many, and it's going to link to the stratigraphic unit. So if you need to create a stratigraphic unit you can find out which number you want, and you can say okay, I need all red umm chronotypes of KWCH. So you can query that from this and it's going to find the SU and it's going to find all the, that, those items. So then we have the catalogue, which is going to be this form, which is going to combine the data, the arte- the SU and the ceramic number, and then they can fill in this data for the catalogue item. Let me go back here. Umm so those are the, I think I covered them all, excavation unit, I just broke that out into its separate table, uhh to keep it umm normalized.

Zack: Yeah.

Greg: So I created all the fields, I created the umm, let's go into one of 'em. Go to the structure. So here you can see umm the ceramic ID, umm it's tied there, SU is a foreign key. So SU is going to tie to the SU table.

Zack: Yeah.

Greg: So that's what the tables look like. And if you look at the others, they're similar. Umm. So here's the catalogue number, you have the ceramic ID number, umm labels, scan, photo, and all this information, umm. That's going to tie to that. So, oh wait, I need to put the SU field in here too. Umm so that's that stuff. Where I'm at now is created the forms. So I'm creating an input form where you can put the data into the database, where you'll go you'll say enter new, there would be a main page, say enter new items, enter new EU, area, whatnot.

Zack: Yeah.

Greg: And then they'll go through, his next, and it would update.

B6

[77] **Zack:** Yeah. One thing I've also noticed, as someone who does database work, a lot of these technical aspects are hard to claim authorship for.

Rufus: Sure.

Zack: I've had this experience recently where I was a little bit peeved about having done so much work, not having been credited in a publication. I'm wondering if that's something that you see as potentially changing

in the field.

Rufus: Maybe I guess from the director... I was in your exact position, I was running databases and GIS and everything, but I was always given the opportunity to, if I wanted it, to publish stuff, always. So with the photo— the JFA article and all those other articles, I was given the opportunity to do that, provided that I took an active role in that. For for [redacted project name], example, yeah, any report or report [unclear] our department of antiquities publications or whatever, as a trench supervisor student, I was never put on reports. I was never put on those publications. Once it became field director, which is obviously under co-director, I was. And I was given opportunities to take, to be a lead on these things. I understood that from their perspective that they are bringing over people like trench supervisors, bringing them over, paying their way, room and board, all that stuff, and they had a specific task to do. When they did that task, that information is then absorbed into the project. If they went above and beyond or wanted to take an active role in something, there was opportunities to do that.

Zack: Specifically regarding that, so I I've been trying to find ways of like representing my work, like digital work, which is often like very specific to a specific locus or specific context of work, right. So how did, like it's like, so how do you share this, this collation of data, [unclear] learned a lot, you know these skills that other people don't have, you share that information, that's the sort of thing that I've been struggling with. As these sorts of things become more pervasive, how do you think these are going to be handled?

Rufus: Yeah, I don't know.

Zack: That's a strange question, I haven't really thought through that.

Rufus: Yeah. And I don't know how they would change necessarily. Because on a project director side, this is a 365 day a year deal where you're always writing grants or writing publications, thinking about the project, game planning for next season, getting resources together. And there's a tremendous investment in that. Versus, say, a trench supervisor that's rolling in for three weeks writing the report and then handing it in. That level of contribution wouldn't necessarily equate to an authorship on a peer reviewed publication. And granted, I was that person—

Zack: I'm not trying to pass judgement, I'm just trying to get your perspective.

Rufus: And that would be my perspective I guess. Yeah. If a graduate student or the, I don't really see undergraduates in that particular position. But if that trench supervisor is like, hey, if it, in my instance, if I had somebody come up to me who was a trench supervisor and say, you know, I'm really interested in this thing, I would foster that interest. They could take that on. And that's something that we can authorship on give an the publication. For the creation of a report, no, because what the person is putting in that report is discussions that I had with them. Umm, they're working within a rigid framework of report format style that I gave to them. And it's more of a, it becomes more of a job than an academic slash scholarly contribution. So that's kind of where I see. But, granted again, I was, I was in a trench supervisor position in that exact same scenario, and I guess I never really gave it a thought because I knew that, you know, the sling bullets here, I was, that's something that I was in. Like hey, I'm interested interested in this. really And then they're like, all right, well, go ahead. And they didn't even want their names on it. I was able to publish it all on my own. And so, I am keen to provide opportunities like that for people. But yeah, when it comes down to that all-important peer reviewed publication, having three names on it is very different than having 15 names on it.

B7

[78] **Zack:** Would you, but is there more room, like so, the specialists are not as much involved in that process of synthesizing for that, then. You just take the key conclusions and...

Rufus: I take the conclusions, so if I was doing it, I would take the conclusions and I'd work them into the overarching argument.

Zack: Yes, can you just explain the overall process to me, maybe make it more clear. Yeah.

Rufus: Yeah, exactly.

Zack: Yeah. No, I mean...

Rufus: Oh, you're asking me to do that now? Okay. So the overall process being alright, department of antiquity report, whatever specialist report comes in, that's immediately appended verbatim into the final report, with on their name it. When it comes time to actually do a peer reviewed publication, there's a lead author. That lead author has access to that document that they can cut, paste, amend, bring in, rework, whatever they want to do, including those specialist reports. Once that document is then complete, the other co-director gets it to read it over to make their changes. Once that's done, that then goes to the specialist to look over their stuff, to make sure something wasn't misconstrued or whatever. They will then fix, amend as necessary. Then the primary author goes through it all over again. As far as pictures and images go. Those are then created, and then those who made a contribution to the paper, they are listed as authors.

B8

[79] **Zack:** Are any there potential areas that will, like that you think you may, that you might, that you may have a big role in be producing?

Liz: Sort of like spearhead?

Zack: Yeah.

Liz: It's both the blessing and curse of having directors that are really talented, in that they take control in a lot of these things.

Zack: I guess it's also, like you're doing some things that are very similar to what other people do in that area.

Zack: Right. Yeah. So my skill set overlaps almost 100 percent with Rufus's, except for my technological abilities are less. Haha, so there's, you know, my, I'm sort of like a redundant person.

Zack: Well, I wouldn't go that far.

Zack: Well, you know what I mean. So I think, or what I guess what I hope to sort of spearhead in the future is processing and querying data once it's in the database. That's my goal in the future, in a way that maybe the directors are less interested in doing. And I think that's because my own dissertation is all about querying datasets from across sites and figuring out patterns. And so I want to do that here.

B9

[80] **Zack:** I noticed there's lots of people overlapping.

Rufus: Yeah, that's on purpose. Yup. I mean we have a site that can make a unique contribution to the Hellenistic period and our understanding of it, and it's better to have ten brains to do that than one. Umm-

Zack: Do you find there's more agreement or disagreement because of it?

Rufus: I find that there's more agreement. I mean the, I'm a laid back guy. Uhh we have no room for drama on the project, or you know, competitiveness or conflict or whatever, I shut it down immediately, every single time.

Zack: Yeah, I noticed that this is a very calm project.

Rufus: And that's, that's [unclear]. Yeah, it's better for- it's just way better for peer co- there's a pretty big cognitive process here. You're creating data that didn't exist before. You're making interpretations to create more knowledge that wasn't there before. And uhh it's better to do that in an environment where

people are enjoying being around each other. Umm that people have certain views towards something that you have and whether those views are for or against it, either way that's a good thing. Umm and so those are the values that we extend in the project. Collaborative environment.

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Competing Interests

[83] The author states no conflicts of interest.

Author Contributions

[84] Zachary Batist is the sole author of this work. He defined the scope of the study and identified suitable cases for inclusion, collected and processed all data, performed analysis, interpreted the findings, created all the figures, and wrote the paper.

Informed Consent

[85] Informed consent has been obtained from all individuals included in this study, in compliance with the University of Toronto’s Social Sciences, Humanities, and Education Research Ethics Board, Protocol 34526.

Data Availability Statement

[86] The data generated and analyzed during the current study are included in this published article’s supplementary files.

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